



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to JASMINE ENTERPRISES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

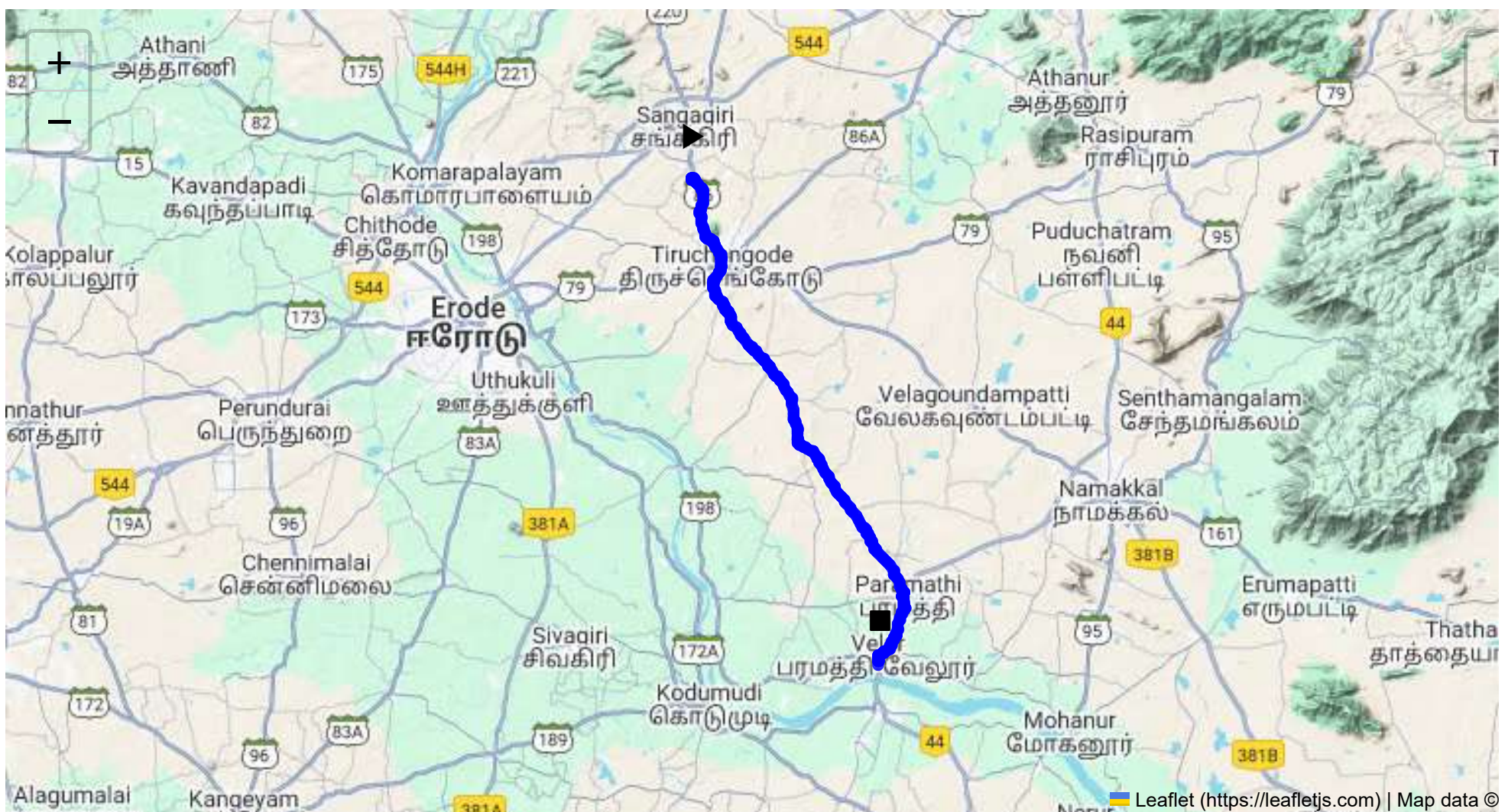
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.27 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.105966, 78.004547)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Velur, Tamil Nadu

1. Overview of the Route Map:

- The route between Sangagiri and Velur covers approximately 44.27 kilometers. It typically involves major highways and state roads connecting these towns. The journey traverses through rural landscapes and small towns, generally following a north-east direction.

2. Typical Weather Conditions and Hazards:

- The region experiences a tropical climate with distinct dry and wet seasons. Monsoon rains (June to September) can cause flooding and slippery roads, while the hot months (March to May) may lead to road surface expansion. Fog is less common but can occur during winter mornings (November to January).

3. Traffic Patterns and Congestion:

- Major congestion is typically observed in the early morning (7:00-9:00 AM) and late afternoon to evening (5:00-7:00 PM) due to local commutes. Key congestion areas include town centers and market areas, particularly around major junctions.

4. Road Quality and Infrastructure Assessment:

- The main highways are generally well-maintained, but secondary roads may exhibit potholes and narrower lanes. Signage is usually adequate but varies in rural sections. Be cautious of unmarked speed bumps within villages.

5. Suggestions for Alternative Routes:

- In case of emergencies or blockages, consider using peripheral roads passing through neighboring smaller villages, although these may increase travel time due to rough surfaces.

6. Summary of Local Regulations for Hazardous Material Transport:

- Transportation of hazardous materials requires compliance with local regulations such as designated routes and time restrictions to avoid peak hours in dense areas. Ensure compliance with labeling and safety equipment standards.

7. Historical Incidents:

- There have been occasional reports of vehicle skidding during the monsoon. A few minor traffic accidents involve heavy vehicles, mostly due to road conditions and driver fatigue.

8. Environmental Considerations and Sensitive Areas:

- The route passes through agricultural areas. Caution is needed to avoid contaminating fields and water bodies. Minimize travel through populated village areas to reduce noise and air pollution.

9. Communication Coverage:

- Generally, the route has good mobile network coverage, though there may be short stretches, especially in rural sections, where signal strength reduces. Network availability can be patchy during inclement weather conditions.

10. Estimated Emergency Response Times:

- Emergency response times vary. In town centers, assistance can typically reach within 20 minutes, whereas more isolated stretches may require 30 to 45 minutes. Emergency services are more concentrated around larger towns and highway intersections.

11. Overall Risk Assessment Summary:

- The route has a moderate risk level. The biggest concerns are weather impacts during monsoon, variable road conditions in certain patches, and traffic congestion in towns. Strict adherence to safety protocols for hazardous materials is essential to mitigate risks. Regular updates on

weather and traffic, alongside scheduled vehicle checks, are advisable before commencing the trip.

This comprehensive analysis aims to familiarize the truck drivers with the potential challenges and risk factors involved in the route to ensure safety and compliance during transportation.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16764, 78.01565	30 KM/Hr
8	Turn	High	11.15573, 78.02176	15 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14480, 78.02291	15 KM/Hr
11	Turn	High	11.12555, 78.01664	15 KM/Hr
12	Turn	High	11.12540, 78.01613	15 KM/Hr
13	Turn	Medium	11.10777, 78.00425	30 KM/Hr
14	Turn	Medium	11.10772, 78.00425	30 KM/Hr
15	Turn	Medium	11.10768, 78.00422	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Indira Hospital, Namakkal	11.1119586, 78.0050232	30 km/h	Medium
20	hospital	Aachi Hospital	11.11017, 78.007066	30 km/h	Medium
21	hospital	Sabari Eye Hospital	11.1079081, 78.0059016	30 km/h	Medium
22	hospital	Sugam Hospital, Karur	11.1066408, 78.0050459	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
19	school	KKHSS	11.1108714, 78.0052766	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.12555, 78.01664



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.12540, 78.01613



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.10777, 78.00425



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.10772, 78.00425



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.10768, 78.00422
